Homework 3
Introduction to Complex Analysis
(MATH 4230-40), Spring 2022

Section 1.5 - Basic Properties of Analytic Functions

1. On what sets are the following functions analytic? Compute the derivative for each.

   (a) z^n for $n \in \mathbb{Z}$

   (b) $\dfrac{1}{\left(z+\frac{1}{z}\right)^2}$

   (c) $\dfrac{z}{z^n-2}$ for $n > 0$, $n \in \mathbb{Z}$

2. Prove that $\dfrac{df^{-1}}{dw} = \dfrac{1}{f'(z)}$ where $w = f(z)$ by differentiating $f^{-1}\big(f(z)\big) = z$ using the chain rule. Assume that f^{-1} is defined and is analytic.

3. Use the inverse function theorem to show that if $f : A \to \mathbb{C}$ is analytic and $f'(z) \neq 0$ for all $z \in A$, then f maps open sets in A to open sets.

4. Define $\dfrac{\partial f}{\partial \bar{z}}$ by $\dfrac{\partial f}{\partial \bar{z}} = \dfrac{1}{2}\left(\dfrac{\partial f}{\partial x} - \dfrac{1}{i}\dfrac{\partial f}{\partial y}\right)$. Show that the Cauchy-Riemann equations are equivalent to $\dfrac{\partial f}{\partial \bar{z}} = 0$.

5. On what sets are each of the following functions harmonic?

   (a) $u(x,y) = \mathrm{Im}\left(z + \dfrac{1}{z}\right)$ (where $z = x + iy$)

   (b) $u(x,y) = \dfrac{y}{(x-1)^2 + y^2}$